

Discrete Mathematics I Test 5 - 28.05.2026

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .
- (b) Determine $d_{\min}$. (*Hint: compute the weights of the four generator rows and a few of their sums.*)
- (c) How many errors does this code detect? How many does it correct?
- (d) Is this code perfect?
- (e) The word $\mathbf{r} = (1, 0, 0, 1, 0, 1, 0, 0)$ was received. Assuming at most one transmission error occurred, use syndrome decoding to recover the sent codeword.

0.3 Solution

The code is defined by the generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Since G is already in systematic form $[I_4 \mid P]$, this is a $[8, 4]$ binary linear code with

$$P = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

(a) Parity-check matrix

For a systematic generator matrix $G = [I_k \mid P]$, the parity-check matrix is $H = [P^\top \mid I_{n-k}]$. Here $n - k = 4$ and

$$P^\top = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix},$$

so

$$H = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Verification. We check $GH^\top = 0$ by computing the four inner products of each generator row with each row of H . As a sample: the first row of G is $(1, 0, 0, 0, 0, 1, 1, 1)$ and the first row of H is $(0, 1, 1, 1, 1, 0, 0, 0)$, giving inner product $1 \cdot 0 + 0 \cdot 1 + \cdots + 1 \cdot 0 + 1 \cdot 0 + 1 \cdot 0 = 0$ over $\mathbb{F}_2$. All sixteen products vanish by the systematic construction.

(b) Minimum distance

Because $\mathcal{C}$ is linear, $d_{\min} = \min_{\mathbf{0} \neq \mathbf{c} \in \mathcal{C}} \text{wt}(\mathbf{c})$.

Lower bound. Each column of H has weight 3 (three ones among four entries). By the *column criterion*, $d_{\min} > d$ if and only if every set of d columns of H is linearly independent. Any single column is nonzero, so $d_{\min} \geq 2$. Any two distinct columns differ in at least one row (they are not equal), so no two columns are identical; thus any two columns are linearly independent over $\mathbb{F}_2$ and $d_{\min} \geq 3$. To check $d_{\min} \geq 4$: we would need every three columns to be linearly independent. The first three columns of H are $(0, 1, 1, 1)^\top$, $(1, 0, 1, 1)^\top$, $(1, 1, 0, 1)^\top$; their sum over $\mathbb{F}_2$ is $(0, 0, 0, 1)^\top \neq \mathbf{0}$, so these three are independent. One can verify all $\binom{8}{3} = 56$ triples are independent, giving $d_{\min} \geq 4$. However, taking the first and second generator rows (messages 1000 and 0100) and their sum (message 1100):

$$\mathbf{g}_1 = (1, 0, 0, 0, 0, 1, 1, 1), \quad \mathbf{g}_2 = (0, 1, 0, 0, 1, 0, 1, 1), \quad \mathbf{g}_1 + \mathbf{g}_2 = (1, 1, 0, 0, 1, 1, 0, 0),$$

and $\text{wt}(\mathbf{g}_1 + \mathbf{g}_2) = 4$.

All generator rows have weight 4 (each has one 1 in the identity block and three 1s in P). Any sum of two distinct rows has weight at least 4 as well. The full weight distribution (obtainable by listing all $2^4 = 16$ codewords) is:

$$A_0 = 1, \quad A_4 = 14, \quad A_8 = 1,$$

so every nonzero codeword has weight 4 or 8. Therefore

$$d_{\min} = 4.$$

(c) Error detection and correction

Solution. With $d_{\min} = 4$:

- **Detects** up to $d_{\min} - 1 = 3$ errors.
- **Corrects** up to $t = \lfloor (4 - 1)/2 \rfloor = 1$ error.

(d) Perfectness

A $[8, 4, 4]$ code corrects $t = 1$ error. The Hamming-ball volume is

$$V(8, 1) = \binom{8}{0} + \binom{8}{1} = 1 + 8 = 9.$$

The Hamming bound requires $2^k \cdot V(n, t) = 2^n$ for a perfect code, i.e.

$$2^4 \cdot 9 = 144 \neq 256 = 2^8.$$

This code is not perfect.

Remark. This code is the *extended Hamming* $[8, 4, 4]$ code, obtained from the $[7, 4, 3]$ Hamming code by appending an overall parity-check bit. Extending always increases $d_{\min}$ by 1 but destroys perfectness.

(e) Syndrome decoding of $\mathbf{r} = (1, 0, 0, 1, 0, 1, 0, 0)$

Step 1: Compute the syndrome.

$$\mathbf{s} = H\mathbf{r}^\top = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 \end{pmatrix} \pmod{2} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

(Computing each entry: $s_1 = r_2 + r_3 + r_4 + r_5 = 0 + 0 + 1 + 0 = 1$; $s_2 = r_1 + r_3 + r_4 + r_6 = 1 + 0 + 1 + 1 = 1$; $s_3 = r_1 + r_2 + r_4 + r_7 = 1 + 0 + 1 + 0 = 0$; $s_4 = r_1 + r_2 + r_3 + r_8 = 1 + 0 + 0 + 0 = 1$.)

Step 2: Identify the error position. We seek the column j of H such that $\mathbf{h}_j = \mathbf{s} = (1, 1, 0, 1)^\top$. Reading off the columns of H :

$$\mathbf{h}_1 = (0, 1, 1, 1)^\top, \quad \mathbf{h}_2 = (1, 0, 1, 1)^\top, \quad \mathbf{h}_3 = (1, 1, 0, 1)^\top, \quad \dots$$

Column 3 equals $\mathbf{s}$. **The error is in position 3.**

Step 3: Correct and decode. Flip bit 3 of $\mathbf{r}$:

$$\hat{\mathbf{c}} = (1, 0, \mathbf{1}, 1, 0, 1, 0, 0).$$

Since the code is systematic ($[I_4 \mid P]$), the original message is the first four bits:

$$\mathbf{u} = (1, 0, 1, 1).$$

Verification. Encoding $\mathbf{u} = (1, 0, 1, 1)$:

$$\mathbf{u}G = (1, 0, 1, 1) \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} = (1, 0, 1, 1, 0, 1, 0, 0) = \hat{\mathbf{c}} \checkmark$$